

More on Step Response and Pendulum Modeling

EGR 345

GVSU

Outline

- some thoughts on lab
- review: MSD step response
- more pendulum modeling

Dialing in the right difficulty

- why can't I just give you step-by-step instructions?
 - what happens if I make it too easy?
- what happens if I make it too hard?

Blame former students

- trying to write detail instructions didn't go well for me...

You can always ask me questions

- it is ok to interrupt “off topic” conversations
- how do I know if you are struggling?
- I will never yell at you for asking a questions
 - worst case, I ask you questions and point you in a direction to get your own answers

Complaint from last year

- We were all struggling with lab while he was joking around and playing chess on his phone...
- if I think you are all working hard and no one is asking me any questions...
- I have never delayed to answer a question.

Lab is a work in progress

- rearranging the lectures to focus on MSD systems isn't too hard
- I don't have the perfect MSD experiments yet
 - but you can help with that. . .

Lab Plans

- investigate modeling the pendulum as an MSD system
 - initial condition response and tap input (last week)
 - voltage time domain inputs (pulse and step)
 - Bode system id
- pendulum vibration suppression
- MSD system design project

Lab Conclusion

- If you don't ask me questions, I won't know you are struggling.

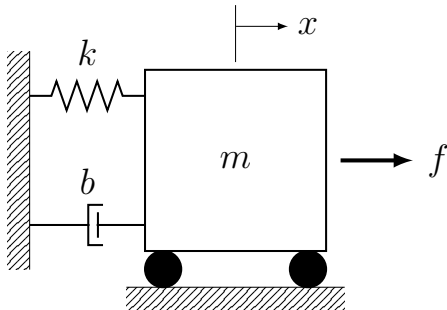
Review

MSD Review

Find the unit step response for an MSD system using the parameters:

- $m = 2$
- $k = 75$
- $b = 1$

verify your answer with Python

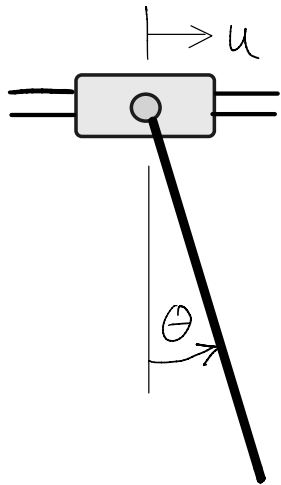


Pendulum Modeling

Is a pendulum basically the same as a mass/spring/damper system?

- what did you learn in last week's lab?
 - initial condition response
 - impulse response from tap/whack
- what about voltage input to the motor on the cart?

Is a pendulum basically the same as a mass/spring/damper system?



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